Key
1 Trim upper right corner only fo r left insert layers A and B.
2 Trim upper left corner only fo r right insert layers A and B.
3 Outside
4 Inside

Insert layer (N)	Buoyancy (approx.)	Insert layer dimensions (mm) a	b	c	d
A	2.8	145	305	30	30
B	2.7	140	300	30	30
C	2.3	125	285	0	0
D	2.0	115	275	0	0
E	1.6	95	255	0	0

Figure 1 – Front foam insert specifications

Insert layer	Buoyancy (approx.) (N)	Insert layer dimensions (mm) a	b	c	R
A	2.8	343	140	9	46
B	2.6	335	133	5	50
C	2.2	325	120	3	52
D	1.9	315	108	0	55
E	1.6	305	95	-5*	55

* measured in direction opposite to that shown in figure.

Figure 2 – Back foam insert specifications
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1 Any thickness of foam up to 7 mm is acceptable for a make-up layer.
2 For 7 mm thick foam, 15,300 mm 2 surface area equals approximately 1 N of buoyancy.

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1 Any thickness of foam up to 7 mm is acceptable for a make-up layer.
2 For 7 mm thick foam, 15,300 mm 2 surface area equals approximately 1 N of buoyancy.

Make-up layer dimensions (mm) Make-up layer (1) Buoyancy (approx.)
(N) Length (mm) (2) Width (mm)
1.0 51 Front 1.5 300 76
1.0 45 Back 1.5 340 67
(1) For 7 mm thick foam.
(2) The length for make-up layer is fixed to maintain proper placement within the lifejacket, but the width may vary to obtain desired buoyancy.

Figure 3 – Child RTD “make-up” foam insert sizes

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APPENDIX 3

INFANT REFERENCE TEST DEVICE (RTD)
DESIGN AND CONSTRUCTION

1 General. The RTD is intended for use only as a test reference standard to represent the desired level of in-water pe rformance of a lifejacket required by the 1974 SOLAS Convention, and is not considered representative of any other required

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1 General. The RTD is intended for use only as a test reference standard to represent the desired level of in-water pe rformance of a lifejacket required by the 1974 SOLAS Convention, and is not considered representative of any other required lifejacket performance. The infant RTD is for persons weighing less than 15 kg, or less than 100 cm in height. The device is designed to fit persons with a chest size of less than 50 cm. This RTD is made with layers of buoyant foam in a bib-style design using a heavy nylon shell cover fabric secured to the body with a waist belt with quick and positive closure and adjustment, along with a chest strap at the neck for closure and adjustment. The shell is made with slide fa steners (zippers) in place of closing seams to hold the foam within, in order that the foam inserts can be easily removed to check their

buoyancy and renew or supplement them if th ey are out of tolerance. The RTD is designed to be reasonably comfortable to wear as a non-reversible device.

2 Materials. All materials used should comply with ISO 12402-7.

2.1 Foam requirements. The performance of the RTD is dependent on using plastic

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designed to be reasonably comfortable to wear as a non-reversible device.

2 Materials. All materials used should comply with ISO 12402-7.

2.1 Foam requirements. The performance of the RTD is dependent on using plastic
foam of the proper stiffness, shapes, and buoyancy.

2.1.1 Stiffness. The buoyant inserts are made of layers of medium stiffness foam to create a flexible but firm buoyancy element.

2.1.2 Shape. The shape of each foam layer is identified in figures C.2 and C.3. Dimensions are in tables C.1, C.2 and C.3.

2.1.3 Buoyancy. The total design buoyancy of the device is 71 N. Table C.4 identifies foam characteristics, the buoyancy for each in sert and its toleranc es, and the overall buoyancy distribution to be verified when using the RTD for certification testing.

2.2 Other component requirements. See table C.1.

3 Construction. The construction and assembly of the device should be in accordance with tables C.1 and C.5 and figures C.1 to C.9. A tolerance of ± 6 mm is used throughout for fabric cutting and stitching assembly. A tolerance of ± 6 mm is also used

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accordance with tables C.1 and C.5 and figures C.1 to C.9. A tolerance of ± 6 mm is used throughout for fabric cutting and stitching assembly. A tolerance of ± 6 mm is also used
for foam cutting, however, the buoyancy requirements of table C.4 should be met.

3.1 Seams. Seam allowances are 13 mm, unless othe rwise specified. All structural seams use a lock type stitch so that the seam will not unravel when a force is applied in the direction of the seam on any of the threads forming the stitch. Stitching should have a density of 7 - 12 stitches per 25 mm of stitch length. Box-x stitching on the webbing is 30 x 15 mm for the waist belt and 15 x 13 mm for the belt loop and chest strap, unless otherwise specified. The bar-tack stitchi ng on webbing is 30 x 2 mm for the waist belt and 15 x 2 mm for the belt loop and chest strap.

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3.1.1 The fabric reinforcements for the waist belt, belt loop and chest strap should be

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3.1.1 The fabric reinforcements for the waist belt, belt loop and chest strap should be attached to the inside surface of the outside cover before attaching any of these items. On the closing seam of the top and bottom secti ons of the outside and inside cover, the cut ends of the fabric are turned under and stitched when instal ling the zippers so that the fabric will not ravel and so that the folds are flush with the line where the zipper teeth mesh (zippers installed to be hidden by cover fabric when closed).